

NANDAN SRIRANGA

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SUMMARY

PhD Candidate in Electrical Engineering specializing in statistical signal processing, sequential inference, distributed detection, optimization, inter-sensor collaboration, and networked cyber-physical systems. Experienced in developing mathematical models, detection algorithms, and MATLAB simulations for quickest change detection, parameter estimation, anomaly detection, and communication-constrained inference in distributed systems.

Strong background in probability theory, statistics, optimization, numerical linear algebra, random processes, and machine learning, with interest in extending inference and optimization methods to critical infrastructure networks and cyber-physical systems.

TECHNICAL SKILLS

Programming & Simulation: Python, MATLAB, C++

Applied Mathematics: Optimization, Linear Algebra, Probability and Statistics, Random Processes

Statistical Inference: Detection Theory, Estimation Theory, Sequential Analysis, Quickest Change Detection, Universal Hypothesis Testing

Networked Systems: Distributed Detection, Sensor Networks, Graph/Topology-Constrained Optimization, Cyber-Physical Monitoring, Passive Radar

Machine Learning: Classification, Regression, CNNs, SVMs, Feature Extraction, PCA, Monte-Carlo Tree Search

EDUCATION

PhD Candidate in Electrical Engineering, Syracuse University Aug 2019–Present

GPA: 3.83/4.0 (52 credits)

Expected Graduation Date: October 2026

B.E. Electronics and Communications Engineering, PES University, Bangalore, India 2014–2018

GPA: 9.43/10

RESEARCH FOCUS

Sequential inference, quickest change detection, sparse and heterogeneous network outage detection, distributed detection, topology-constrained optimization, universal hypothesis testing, and communication-constrained inference for networked cyber-physical systems.

PUBLICATIONS

In Preparation/Under Peer Review

1. **Nandan Sriranga**, A. Grootveld, and P. K. Varshney, “Online sensor collaboration for the detection of a moving target in distributed sensor networks,” **under review** by IEEE Transactions on Signal Processing, 2026.
2. **Nandan Sriranga** and P. K. Varshney, “Sparsity-aware penalized Rao-score based statistics for quickest detection of heterogeneous link outages,” **in preparation** for submission to IEEE Transactions on Signal Processing.
3. **Nandan Sriranga**, A. Grootveld, and P. K. Varshney, “Sequential Universal Two-Sample Testing Using Predictive Likelihoods with Adaptive and Active-Sensing Policies,” **in preparation** for submission to IEEE Transactions on Information Theory.

Accepted Papers

1. **Nandan Sriranga**, H. Yang, and P. K. Varshney, "Quickest detection of link outages in wireless sensor networks using a locally most-powerful statistic," accepted for presentation at IEEE International Conference on Information Fusion (FUSION), Trondheim, Norway, 2026.
2. **Nandan Sriranga**, H. Yang, V. Gandikota, and P. K. Varshney, "Target detection in two-channel passive radars with inter-receiver collaboration," in IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Barcelona, Spain, 2026.
3. A. Grootveld, H. Yang, **Nandan Sriranga**, B. Chen, V. Gandikota, and J. Pollack, "Asymptotically optimal quantum universal quickest change detection," IEEE International Symposium on Information Theory (ISIT), Guangzhou, China, 2026.
4. **Nandan Sriranga** and P. K. Varshney, "Covert communication in distributed multi-user wireless networks using inter-user collaboration," 2025 59th Asilomar Conference on Signals, Systems, and Computers, 2025.
5. **Nandan Sriranga**, A. Grootveld, and P. K. Varshney, "Linear sensor collaboration for distributed parameter estimation in the presence of communication failures," in 2025 28th International Conference on Information Fusion (FUSION), pp. 1–8, IEEE, 2025.
6. C. Quan, S. Bulusu, B. Geng, Y. S. Han, **Nandan Sriranga**, and P. K. Varshney, "On ordered transmission based distributed Gaussian shift-in-mean detection under Byzantine attacks," IEEE Transactions on Signal Processing, vol. 71, pp. 3343–3356, 2023.
7. C. Quan, **Nandan Sriranga**, H. Yang, Y. S. Han, B. Geng, and P. K. Varshney, "Efficient ordered-transmission based distributed detection under data falsification attacks," IEEE Signal Processing Letters, vol. 30, pp. 145–149, 2023.
8. **Nandan Sriranga**, A. Trezza, S. Bulusu, D. J. Bucci, and P. K. Varshney, "Online identification of recurring changepoints," in 2022 56th Asilomar Conference on Signals, Systems, and Computers, pp. 426–430, IEEE, 2022.
9. **Nandan Sriranga**, B. Geng, and P. K. Varshney, "On human assisted decision making for machines using correlated observations," in 2020 54th Asilomar Conference on Signals, Systems, and Computers, pp. 1502–1506, IEEE, 2020.
10. **Nandan Sriranga**, C. R. Murthy, and V. Aggarwal, "A method to improve consensus averaging using quantized ADMM," in 2019 IEEE International Symposium on Information Theory (ISIT), pp. 507–511, IEEE, 2019.
11. **Nandan Sriranga**, K. G. Nagananda, and R. S. Blum, "Shared channel ordered transmissions for energy-efficient distributed signal detection," IEEE Communications Letters, vol. 23, no. 1, pp. 96–99, 2018.
12. **Nandan Sriranga**, K. G. Nagananda, R. S. Blum, A. Saucan, and P. K. Varshney, "Energy-efficient decision fusion for distributed detection in wireless sensor networks," in 2018 21st International Conference on Information Fusion (FUSION), pp. 1541–1547, IEEE, 2018.

COLLABORATIVE NETWORKS, SEQUENTIAL INFERENCE & SIGNAL PROCESSING RESEARCH

PhD Candidate, Syracuse University

Advisor: Prof. Pramod K. Varshney

Quickest Detection of Sparse and Heterogeneous Network Outages

This work develops statistical methods for rapidly detecting failures in networked cyber-physical systems. The original formulation considers a wireless sensor network with a fixed communication

topology, where links may fail probabilistically over a finite transient outage window and the link-failure probability is unknown. More broadly, the framework is motivated by infrastructure networks in which failures, attacks, or disruptions may occur locally across communication links, transmission lines, control pathways, or other networked dependencies. I derived a *locally most-powerful statistic* for the homogeneous **small outage probability** regime (rare outages) and integrated it into a CuSum-style **quickest detection** framework to identify outage onset with low detection delay. The statistic enables efficient outage detection by reducing computational complexity, avoiding explicit estimation of unknown outage probabilities, and eliminating the need to evaluate probability mixtures over all possible outage scenarios.

I am now extending this work using **penalized Rao-based statistics** for sparse and heterogeneous outage models, including elastic-net, graph fused lasso, and group-lasso penalties, where only a subset of nodes or links are susceptible to outages and different regions of the network may have different failure probabilities or outage patterns. This extension avoids an exhaustive search over all possible failure scenarios, reduces reliance on uniform link-outage assumptions, and is naturally applicable to critical infrastructure systems such as electric power grids, natural gas networks, and cyber-physical communication networks.

Sequential Two-Sample Universal Hypothesis Testing

I am developing **two-sample sequential universal hypothesis testing** methods for settings where the probability distributions generating two data streams are unknown and fully specified parametric likelihood models are unavailable. The work focuses on sequential decision-making under distributional uncertainty, including adaptive testing methods that improve distribution estimation and active stream-selection policies that sample the most informative data stream at each round.

I designed truncated sequential tests that adaptively assign larger weights to more informative streams while achieving the optimal error exponent for the two-sample universal testing problem. I also developed a controlled-sensing formulation in which only one of the two streams may be sampled at each round, and designed stream-selection policies that attain the optimal error exponents for this setting. The analysis uses supermartingale theory and convergence arguments to control error probabilities and characterize the stopping behavior under adaptive sampling policies.

Online Sensor Collaboration for Moving Target Detection

This work addresses the problem of detecting a moving target using a distributed network of sensors, where sensors collaborate with one another before transmission to a fusion center. The target signal amplitude, initial position, and velocity are unknown, and only a subset of sensors can transmit post-collaboration measurements to the fusion center over multiple sensing rounds.

I developed an online inter-sensor collaboration framework in which **collaboration weights are adaptively updated** using sequential estimates of the target state. I proposed a **grid-search assisted amplitude plug-in GLRT** detector for **non-cooperative moving-target detection** and derived *asymptotic performance guarantees* showing that the *online design approaches an oracle-aided offline collaboration policy*. The theoretical analysis uses martingale theory and martingale central-limit results to characterize the behavior of the online statistic despite the adaptive, data-driven collaboration design. I validated the framework through simulation experiments involving detection and false-alarm probabilities, asymptotic non-centrality parameter metrics, the effects of collaboration noise, and the behavior of the system as the network link-density is varied.

Passive Radar Target Detection with Inter-Receiver Collaboration

This work studies target detection in distributed two-channel passive radar systems, where each receiver observes a reference channel and a surveillance channel. The illuminator of opportunity waveform is unknown but lies in a known low-dimensional signal subspace, and the system must reduce receiver to fusion-center communication bandwidth. I designed a distributed radar collaboration scheme that applies whitening and subspace transformations before cross-correlating measurements between the reference and surveillance channels.

I then designed inter-receiver collaboration weights so that only a subset of receivers transmit to the

fusion center, improving detection performance under communication constraints. The process of exploiting the unknown illuminator's waveform subspace, cross-correlating the channel measurements and then collaborating across radar receivers significantly reduces the uplink transmission bandwidth, thereby providing an implementable architecture in distributed passive radar systems.

Collaborative Precoding for Covert Communication

This work considers privacy-aware collaborative precoding in distributed multi-user wireless networks in the presence of a passive eavesdropper. Multiple users transmit a common message to a legitimate receiver Bob, while limiting the eavesdropper's ability to detect whether communication is occurring. When distributed users can collaboratively precode a common message, the receiver obtains a more reliable message signal while the eavesdropper's ability to detect communication is limited.

I formulated an optimization problem that balances receiver SNR against eavesdropper detectability under collaboration-topology and power constraints. I derived a closed-form collaboration matrix design using a generalized Rayleigh-quotient formulation. Simulations showed that collaboration between users can improve receiver SNR while reducing eavesdropper detection performance and requiring fewer transmitting users.

MACHINE LEARNING PROJECTS

AI Agent for Progressive Chess

This course project addressed sequential decision-making in progressive chess, a chess variant where the number of moves available to a player increases on every turn. The rapidly expanding decision space makes exhaustive search difficult, motivating the use of search algorithms, heuristics, and learning-based guidance.

I developed an AI agent using Monte Carlo Tree Search with UCT-based exploration and a custom heuristic tailored to quantify the strength of moves in progressive chess. I also designed and trained a convolutional neural network to learn state-value priors, then integrated the network outputs into the MCTS framework. This guided the search toward better chess moves, reduced the complexity of the search, and improved decision efficiency.

Statistical Learning and Feature Design for Image Classification

This course project focused on supervised classification and feature design for handwritten digit recognition using MNIST. The goal was to compare high-dimensional classification using raw pixel features against compact feature representations that improve computational efficiency while preserving classification accuracy.

I developed SVM-based classifiers with RBF kernels, performed hyperparameter tuning using grid search and 5-fold cross-validation. I also designed PCA-based and symmetry-based feature extraction techniques that reduced the feature space from 784 dimensions to 5 dimensions. I analyzed classification accuracy versus efficiency tradeoffs and extended the models to multi-class classification using the one-versus-all method, with around 93% test accuracy.

RELEVANT COURSEWORK

Signal Processing & Inference: Detection & Estimation Theory, Digital Signal Processing, Random Processes, Digital Communications, Information Theory

Mathematics & Optimization: Probability & Statistics, Linear Algebra, Convex Optimization, Statistical Ranking, Selection & Multiple Comparisons, Linear Models

Machine Learning: Introduction to AI, Big Data, Statistical Learning Theory

TEACHING EXPERIENCE

Teaching Assistant, Syracuse University

- Teaching experience in signal processing and systems courses including Linear Signals & Systems, Probability & Statistics, and Digital Design Labs.